

Analysis on Manifolds — Exam Prep

June 8, 2026



Category 1: Integration on Parameterized Manifolds (Surface Area & Integrals)

Relevant Questions:

1. Calculate the $\iint_{\Sigma} y^2 z \, d\sigma$ when Σ is given by $\Sigma := \left\{ (x, y, z) \mid x = \sqrt{3}y + \frac{\sqrt{5}}{2}z^2, -1 \leq y \leq 1, 0 \leq z \leq 1 \right\}$.
2. Calculate the area given by $r(\theta, \varphi) = (\cos(\varphi) \sin(\theta), \sin(\varphi) \sin(\theta), \cos(\theta))$ when $0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \frac{\theta}{2}$

Core Concept: Calculating integrals over a manifold using a local parametrization and the Gram matrix to find the volume/area distortion.

Steps for Question 1 (1):

1. Identify the given parametrization of the surface Σ based on the constraints: $\varphi(y, z) = \left(\sqrt{3}y + \frac{\sqrt{5}}{2}z^2, y, z \right)$.
2. Calculate the tangent vectors by taking the partial derivatives $\partial\varphi/\partial y$ and $\partial\varphi/\partial z$.
3. Compute the Gram matrix $D\varphi^T D\varphi$ and its determinant to find the area element $d\sigma = \sqrt{\det(D\varphi^T D\varphi)} dy dz$.
4. Substitute the parametrization into the integrand so that $f(\varphi(y, z)) = y^2 z$.
5. Evaluate the double integral over the given bounds $-1 \leq y \leq 1$ and $0 \leq z \leq 1$.

Steps for Question 2 (7):

1. Recognize that the given parametrization $r(\theta, \varphi)$ describes a portion of the unit sphere.
2. Calculate the partial derivatives $\partial r/\partial\theta$ and $\partial r/\partial\varphi$.
3. Compute the area element $\sqrt{\det(Dr^T Dr)}$.
4. Set up and evaluate the double integral $\int \int 1 d\sigma$ using the given bounds for θ and φ .

Category 2: Green's Theorem & Planar Line Integrals

Relevant Questions:

1. Calculate the area inside the infinity sign shape which is described by the path $\gamma(t) = (\sin(t), \sin(t) \cos(t))$ when $0 \leq t \leq 2\pi$
2. Calculate the line integral $\int_{\gamma} \vec{F} d\vec{\ell}$ when $\vec{F}(x, y) = (e^{\sin(y)}, xe^{\sin(y)} \cos(y) + 6x)$ and γ in the path the connects between $(0, 0)$ to $(1, 0)$ by the following: from $(0, 0)$ we go up to $(0, 1)$ then left to $(-1, 1)$ then down to $(-1, 0)$ and then by the lower half circle using $x^2 + y^2 = 1$ to go from $(-1, 0)$ to $(1, 0)$
3. Let $\vec{F}(x, t) = (y^2 - y, (x^3 - x) \cos(y^2))$ and let γ the path that describes the boundary of the unit square.
 1. Calculate $\oint_{\gamma} \vec{F} d\vec{\ell}$
 2. Does $\vec{F}$ is Conservative vector field in $\mathbb{R}^2$?

Core Concept: Relating line integrals along a closed curve to double integrals over the enclosed region using Green's Theorem.

Steps for Question 1 (2) (Area via Green's Theorem):

1. Note that the curve $\gamma(t)$ is a closed path (a figure-eight).
2. Use the corollary of Green's Theorem for area: $\text{Area}(\Omega) = \frac{1}{2} \int_{\partial\Omega} x dy - y dx$.
3. Compute $dx = \gamma_{1'}(t) dt$ and $dy = \gamma_{2'}(t) dt$.
4. Substitute these into the formula and integrate over $0 \leq t \leq 2\pi$.

Steps for Question 2 (8) (Path Completion):

1. Observe that the path γ is not closed (it starts at $(0, 0)$ and ends at $(1, 0)$).
2. Close the path by adding a straight line segment L from $(1, 0)$ back to $(0, 0)$ to form a closed curve $\partial\Omega = \gamma \cup L$.
3. Calculate the mixed partials $\partial Q/\partial x - \partial P/\partial y$ for the given vector field.
4. Apply Green's Theorem over the enclosed region Ω .
5. Subtract the line integral over the added segment L from your result to isolate the integral over γ .

Steps for Question 3 (9):

1. To determine if F is conservative (Part b), check if it is locally conservative by verifying if $\partial F_1/\partial y = \partial F_2/\partial x$. If they are unequal, the field is not conservative.
2. For Part a, apply Green's Theorem directly since γ is a closed curve bounding the unit square.
3. Evaluate the double integral of $(\partial Q/\partial x - \partial P/\partial y)$ over the square $[0, 1] \times [0, 1]$.

Category 3: Conservative Vector Fields

Relevant Question:

1. Calculate the line integral $\int_{\gamma} \vec{F} d\vec{\ell}$ of the vector field $\vec{F}(x, y) = (x^2 y + \sin(x) + y \sin(xy), \frac{x^3}{3} + x \sin(xy))$ along the path $\gamma(t) = (t \cos(t), t \sin(t))$ where $\pi \leq t \leq 4\pi$

Core Concept: Line integrals of conservative fields are path-independent and depend only on the endpoints.

Steps for Question 3 (3):

1. Check if the vector field is locally conservative by verifying $\partial F_1/\partial y = \partial F_2/\partial x$.
2. Because the domain is $\mathbb{R}^2$ (which is star-shaped), a locally conservative field is conservative, meaning it has a potential function u such that $\nabla u = F$.
3. Integrate the components of F to find the potential function $u(x, y)$.
4. Evaluate the path endpoints $p = \gamma(\pi)$ and $q = \gamma(4\pi)$.
5. Calculate the line integral simply by taking the difference $u(q) - u(p)$.

Category 4: Divergence Theorem (Gauss's Theorem)**Relevant Questions:**

1. Calculate the flux of the field $\vec{F}(x, y, z) = (-y, x, z)$ over the outside side closed plane Σ where

$$\Sigma := \left((x, y, z) \mid x^2 + y^2 = \cos^2(z), -\frac{\pi}{2} \leq z \leq \frac{\pi}{2} \right)$$

2. Let

$$\Sigma := \left((x, y, z) \mid x^2 + y^2 + z^2 = 1, z > -\sqrt{x^2 + y^2} \right)$$

$$\vec{F}(x, y, z) = \left(\frac{x^2}{2} + xy - 4xz + \cos(z), -\frac{y^2}{2} - xy + x^2 + z^3, 2z^2 + x \right)$$

and calculate the flux of vector field $\vec{F}$ by Σ in up orientation (השטף של השדה דרך Σ כלפי מעלה)

Core Concept: Converting flux integrals over closed surfaces into volume integrals of the divergence.

Steps for Question 1 (4):

1. Verify that Σ forms a completely closed surface (the bounds on z pinch the cylinder shut).
2. Calculate the divergence of the vector field: $\text{div } F = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$.
3. Apply the Divergence Theorem: $\int_{\Sigma} \langle F, N \rangle d\sigma = \int \int \int_{\Omega} \text{div } F dV$.
4. Evaluate the triple integral over the enclosed volume Ω , likely using cylindrical coordinates.

Steps for Question 2 (10) (Surface Completion):

1. Recognize that Σ is an open surface (a "cap" of a surface). Close the volume by adding the conical surface S_{cone} defined by the boundary constraint.
2. Calculate the divergence of F .
3. Apply the Divergence Theorem over the newly closed volume to find the total flux.
4. Explicitly calculate the flux over your added surface S_{cone} and subtract it from the total flux to find the flux exclusively over Σ .

Category 5: Stokes' Theorem**Relevant Questions:**

1. Calculate the line integral $\oint_{\gamma} \vec{F} d\ell$ where $\vec{F}(x, y, z) = (\sin(e^{x^2}) + yz, x \cos(y), xz^2)$ and γ is the path the connected using lines the points $(0, 0, 0) \rightarrow (0, 2, 0) \rightarrow (2, 2, 2) \rightarrow (2, 0, 2) \rightarrow (0, 0, 0)$
2. Calculate the line integral $\oint_{\gamma} \vec{F} d\ell$ when $\vec{F}(x, y, z) = (y, -2z, 4x)$ and γ is a circle in radius 2 that is contained in the plane $x + 2y + 3z = 4$ and clockwise when we look up מגמתי

Core Concept: Relating the line integral around a closed curve to the surface integral of the curl over an enclosed surface.

Steps for Question 1 (5):

1. Verify that the given points form a closed, piecewise smooth path γ .
2. Identify a flat planar surface Σ whose boundary is γ (e.g., notice the relationship between the x and z coordinates of the vertices).
3. Compute the curl of the vector field F .
4. Determine the unit normal vector N of your chosen plane.
5. Apply Stokes' Theorem: $\int_{\gamma} F \cdot dl = \int \int_{\Sigma} \langle \text{curl } F, N \rangle d\sigma$.

Steps for Question 2 (11):

1. Identify the surface Σ as the disk of radius 2 lying on the plane $x + 2y + 3z = 4$.
2. Compute $\text{curl } F$.
3. Extract the normal vector N directly from the plane equation, ensuring it points in the correct direction to match the clockwise orientation of the curve.
4. Apply Stokes' Theorem and evaluate the flux of the curl across the disk.

Category 6: Chain Rule & Differentials

Relevant Question:

1. Let $f(x, y)$ function that all its derivatives from second order are continuous.
Let $\gamma_1(t) = (t, 0)$, $\gamma_2(t) = (t, \frac{t^2}{2})$, $\gamma_3(t) = (\sin(t), 1 - \cos(t))$ and let $g_j(t) = f(\gamma_j(t))$
 1. Foreach $1 \leq i < j \leq 3$ decide if it must happen $g'_i(0) = g'_j(0)$
 2. Foreach $1 \leq i < j \leq 3$ decide if it must happen $g''_i(0) = g''_j(0)$

Core Concept: Applying the chain rule to composite functions along curves.

Steps for Question 1 (6):

1. Evaluate the position $\gamma_j(0)$, velocity $\gamma'_j(0)$, and acceleration $\gamma''_j(0)$ for all three curves.
2. For Part a, use the chain rule to express the first derivative $g'_j(t) = \nabla f(\gamma_j(t)) \cdot \gamma'_j(t)$. Evaluate at $t = 0$ and check if the velocity vectors are identical.
3. For Part b, differentiate again using the product rule and chain rule to express $g''_j(t)$. Evaluate at $t = 0$ to see if identical accelerations result in identical second derivatives.

Category 7: Differential Geometry (Curvature)

Relevant Question:

1. Let Σ given by $z = xy$. Calculate Gaussian curvature K in the origin and the average curvature H in the origin.

Core Concept: Calculating the second fundamental form to determine curvatures.

Steps for Question 1 (12):

1. Write the surface as a parametrized map $\varphi(x, y) = (x, y, xy)$.
2. Compute the unit normal vector field N across the surface.
3. Compute the shape operator (the differential of the normal map) $A_p(v) = DN|_p(v)$. This yields the matrix for the second fundamental form.
4. **External Knowledge Note:** Calculate the determinant of this matrix to find the Gaussian curvature (K).
5. **External Knowledge Note:** Calculate half the trace of this matrix to find the Mean curvature (H).

Integration on Parameterized Manifolds

Objective: Calculate the integral of a scalar function over a manifold, or find its volume/area.

General Method:

- **Parametrize:** Identify or define a local parametrization $\alpha : U \rightarrow M$ for the manifold.
- **Differentiate:** Compute the differential (Jacobian matrix) $D\alpha|_x$.
- **Find the Distortion Factor:** Calculate the volume distortion factor, $V(D\alpha|_x)$, which is given by the square root of the Gram determinant: $\sqrt{\det(D\alpha^T D\alpha)}$.
- **Integrate:** Set up the integral over the parameter domain U : $\int_U f(\alpha(x))V(D\alpha|_x)dx$. Evaluate this as a standard multiple integral.

Green's Theorem & Planar Line Integrals

Objective: Relate a line integral around a closed planar curve to a double integral over the enclosed region.

General Method:

- **Check the Path:** Ensure the curve is simple, closed, and positively oriented. If the path is *not* closed, artificially add a segment to close it, apply the theorem, and subtract the line integral of the added segment at the end.
- **Identify Components:** Extract the vector field components P and Q .
- **Differentiate:** Calculate the partial derivatives $\partial \frac{Q}{\partial x}$ and $\partial \frac{P}{\partial y}$.
- **Integrate:** Evaluate the double integral over the enclosed region Ω : $\int \int_{\Omega} \left(\partial \frac{Q}{\partial x} - \partial \frac{P}{\partial y} \right) dx dy$.

Conservative Vector Fields

Objective: Evaluate line integrals efficiently by finding a scalar potential function.

General Method:

- **Verify Local Conservatism:** Check if the field F is locally conservative by verifying that $\partial \frac{F_j}{\partial x_i} = \partial \frac{F_i}{\partial x_j}$ for all components.
- **Check the Domain:** If the domain is simply connected or star-shaped, a locally conservative field is guaranteed to be globally conservative.
- **Find the Potential:** Find the potential function φ such that $\nabla \varphi = F$ by integrating the components of the field one by one.
- **Apply the FTC:** Use the fundamental theorem for line integrals: $\int_{\gamma} F \cdot dl = \varphi(\gamma(b)) - \varphi(\gamma(a))$.

Divergence Theorem (Gauss's Theorem)

Objective: Convert the flux of a vector field across a closed surface into a volume integral over the enclosed solid.

General Method:

- **Check the Surface:** Ensure the surface $\partial\Omega$ is completely closed (bounds a compact region Ω) and the normal vector points outward. If the surface is open (e.g., a cylinder missing a lid), close it with an additional flat surface, apply the theorem, and subtract the flux through your added surface.
- **Compute Divergence:** Calculate the divergence of the field: $\text{div } F = \sum \partial \frac{F_i}{\partial x_i}$.
- **Integrate:** Evaluate the triple integral over the enclosed volume Ω : $\int \int \int_{\Omega} \text{div } F dV$.

Stokes' Theorem

Objective: Relate the line integral around a closed curve in 3D space to the flux of the curl over a bounding surface.

General Method:

- **Choose a Surface:** Identify a surface Σ whose boundary $\partial\Sigma$ matches the given curve. Often, choosing the simplest, flattest surface makes the math easier.
- **Orient Correctly:** Ensure the unit normal N of the surface and the orientation of the boundary curve T follow the right-hand rule.
- **Compute Curl:** Calculate the curl of the vector field: $\text{curl } X = \left(\partial \frac{R}{\partial y} - \partial \frac{Q}{\partial z}, \partial \frac{P}{\partial z} - \partial \frac{R}{\partial x}, \partial \frac{Q}{\partial x} - \partial \frac{P}{\partial y} \right)$.
- **Integrate:** Evaluate the surface integral: $\int \int_{\Sigma} \langle \text{curl } X, N \rangle d\sigma$.

Chain Rule & Differentials

Objective: Differentiate compositions of multivariable functions.

General Method:

- **Deconstruct:** Identify the constituent functions (e.g., g and f) and their respective domains and codomains.
- **Find Individual Differentials:** Compute the Jacobian (differential) matrix for each function individually: (Dg) and (Df) .
- **Multiply:** Multiply the matrices, ensuring the outer function's derivative is evaluated at the inner function's output: $D(g \circ f)_p = (Dg)_{f(p)} \cdot (Df)_p$.

Differential Geometry (Curvature)

Objective: Calculate the Gaussian or Mean curvature using the shape operator.

General Method:

- **Parametrize and Find the Normal:** Define a local parametrization for the surface and compute the unit normal field N .
- **Compute the Shape Operator:** Compute the differential of the normal map, known as the shape operator: $A_p(v) = DN|_p(v)$.
- **Construct the Matrix:** Build the matrix of this operator, which represents the second fundamental form.
- **Extract Curvatures:** Find the determinant of this matrix for the Gaussian curvature (K), or find half the trace for the Mean curvature (H).